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| Seat No. |

Time: 13.00-16.00

Cheating in the exam is considered an extremely serious offence which will result in expulsion from the University.

This exam paper is part of a set used to assess student abilities as follows:

|       |                                                                                                             | Question |
|-------|-------------------------------------------------------------------------------------------------------------|----------|
| CLO 1 | Explain and use the theory of mathematical functions and trigonometric functions.                           | Q1-Q9    |
| CLO 2 | Evaluate the values of mathematical functions and trigonometric functions.                                  | Q1-Q9    |
| CLO 3 | Understand and analyze mathematical contents related to mathematical functions and trigonometric functions. | Q1-Q9    |
| CLO 4 | Write the steps to solve mathematical problems.                                                             | Q1-Q9    |
| CLO 5 | Use mathematical techniques to solve mathematical problems.                                                 | Q1-Q9    |

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**Question 1.**

1.1 Determine which of the following relations is a function or not. Why?

(a)  $r_1 = \{(a, 1), (b, 2), (c, 3), (d, 1), (a, -1)\}$ . (2 points)

(b)  $r_2 = \{(x, y) \in \mathbb{R} \times \mathbb{R} \mid y = (x - 2)^2 + 4\}$ . (2 points)

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1.2 Given the function  $f = \{(1, 0), (2, 1), (3, 5), (4, 6), (5, 5)\}$ .

(a) Find  $D_f \cap R_f$ . (3 points)

(b) Find the value of  $f(3) - f(2)$ . (3 points)

## Question 2.

Given the functions  $f(x - 3) = 2x - 1$ ,  $g(x) = \frac{x-1}{x^2-4}$ , and  $h(x) = \sqrt{x-1}$ .

(a) Find  $f(x^2)$  and  $f^{-1}(3)$ .

(5 points)

(b) Find  $D_{h-g}$  and  $(h-g)(5)$ .

(5 points)

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(c) Find  $D_{\frac{g}{h}}$  and  $\left(\frac{g}{h}\right)(1)$ .

(3 points)

## Question 3.

3.1 Given the functions  $f(x) = \frac{x+2}{x-1}$  and  $(f^{-1} \circ g)(x) = \frac{x}{x+1}$ .

Find  $g(x)$  and determine whether  $g$  is a decreasing function. (5 points)

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3.2 If  $f(x) = \frac{1}{x}$  and  $g(x) = 3f(x)$ , find the value of  $(f \circ g^{-1})(1)$ . (4 points)

## Question 4.

4.1 Given  $\cos x = \frac{4}{5}$  and  $\sin x < 0$ . Find the value of  $\tan x$ . (3 points)

4.2 Given  $x \in [0, 2\pi]$  and  $x \neq \frac{\pi}{2}, \frac{3\pi}{2}$ .

Show that  $\frac{1}{1 + \sin x} + \frac{1}{1 - \sin x} = 2 \sec^2 x$ . (3 points)

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4.3 Find the value of  $\frac{\cos 150^\circ + \sin(-60^\circ)}{\tan\left(\frac{5\pi}{4}\right)}$ . (3 points)

## Question 5.

5.1 Determine the value of  $\cos 73^\circ \cos 13^\circ + \sin 13^\circ \sin 73^\circ$ . (4 points)

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5.2 If  $\frac{\pi}{2} < \alpha < \pi$ ,  $\pi < \beta < \frac{3\pi}{2}$ ,  $\sin \alpha = \frac{5}{13}$  and  $\cos \beta = -\frac{4}{5}$ , then find the value of  $\tan(\alpha + \beta)$ . (5 points)

## Question 6.

6.1 If  $\sin \theta = -\frac{1}{2}$  and  $\pi < \theta < \frac{3\pi}{2}$ , find the value of  $\sin 2\theta$ . (4 points)

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6.2 Show that  $\cos\left(\frac{\pi}{4} + \theta\right) + \cos\left(\frac{\pi}{4} - \theta\right) = \sqrt{2} \cos \theta$ . (4 points)



## Question 7.

7.1 Show that  $\frac{\cos 6\theta + \cos 4\theta}{\sin 6\theta - \sin 4\theta} = \cot \theta$  .

(4 points)

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7.2 Find the value of  $\sin^{-1}(\sin(\frac{3\pi}{5}))$ .

(3 points)

Question 8.

8.1 Evaluate  $\cos\left(\arcsin\frac{3}{5} + \arcsin\frac{5}{13}\right)$ .

(5 points)

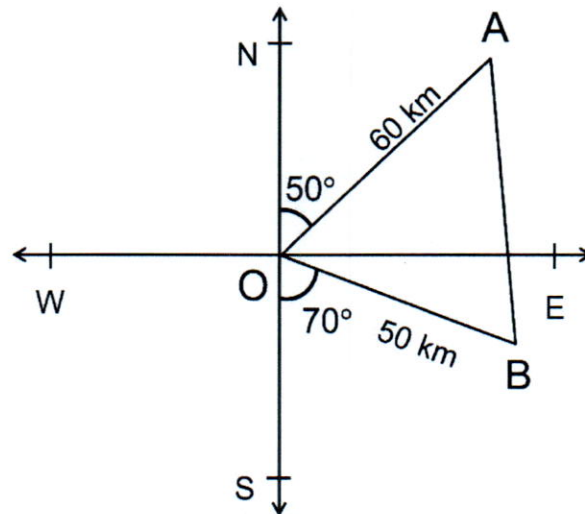
8.2 Solve the equation  $2\cos^2 x - 3\sin x = 0$ ,  $0 \leq x \leq 2\pi$ .

(5 points)

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**Question 9.** Let the original position of two boats be at point  $O$ . Let  $A$  and  $B$  be their respective final positions, as shown in the figure. Find the distance between the boats.

(8 points)



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